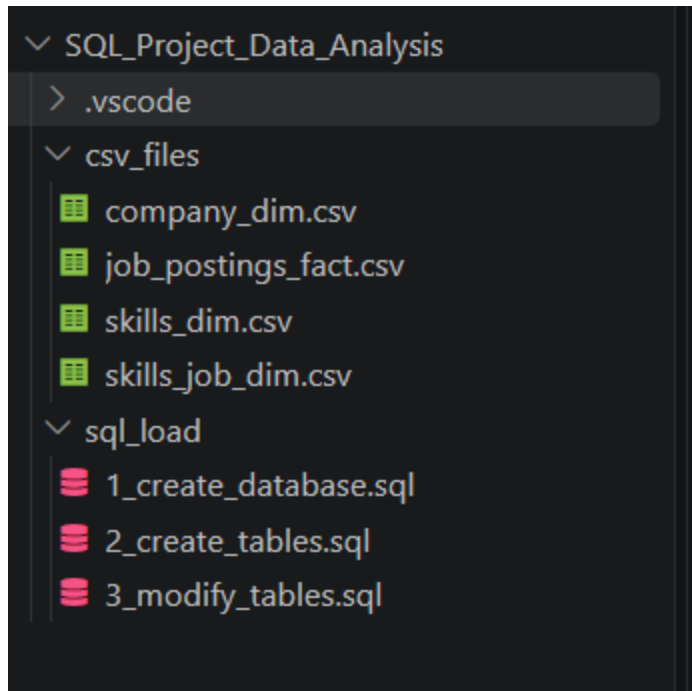


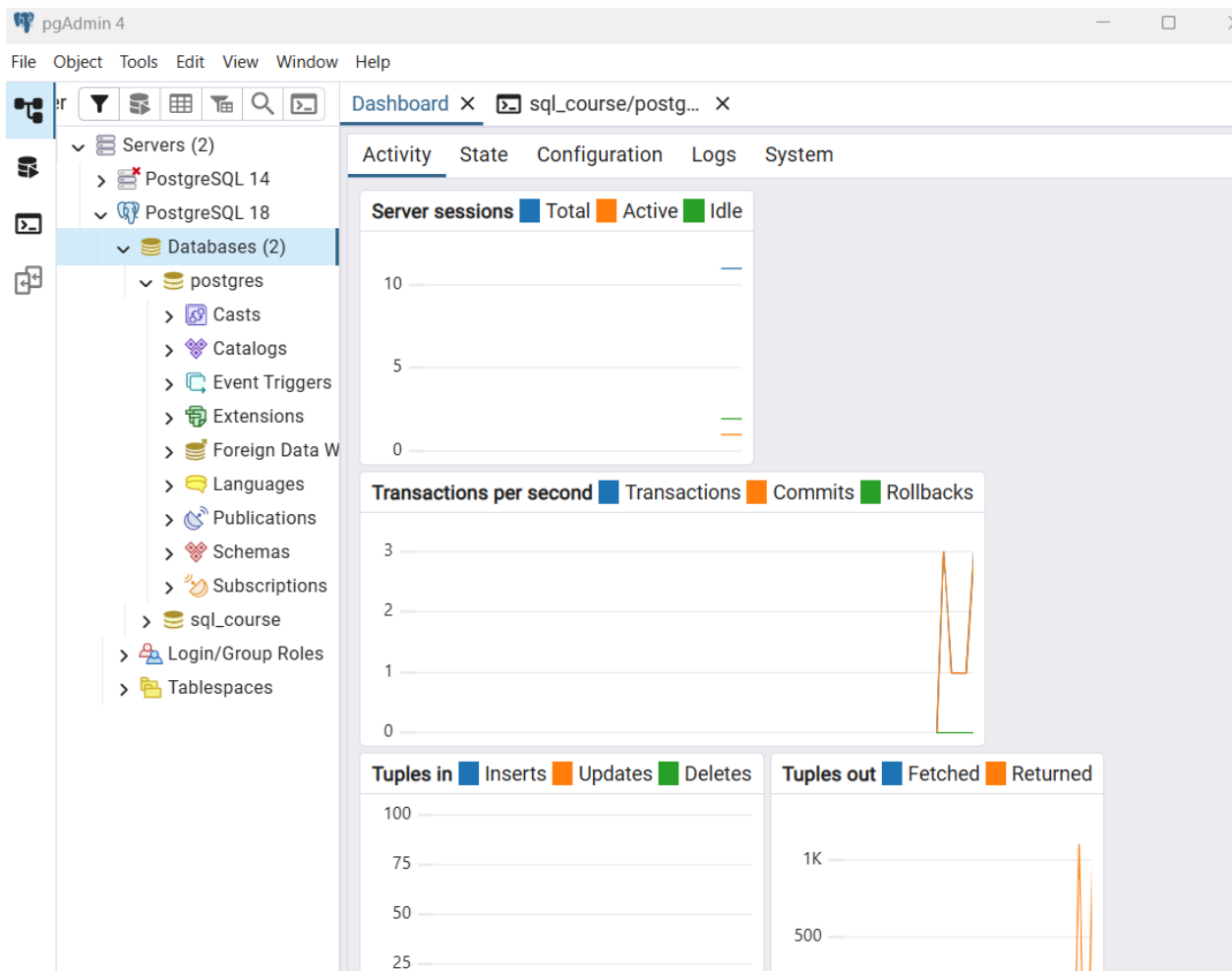


Created the database for the SQL project on VS code using SQL code. Proceeded then to upload two csv files (extracted first) onto the database and SQL project folder onto VS Code, I then also created two different SQL connections on PG ADMIN PostgreSQL 18 named the following connections; postgres and sql_course. I then took the two connections and connected them both onto the vs code project database connectors using SQL Tools extension to connect the two. I tested that the SQL databases ran with the connections by creating databases and tables for each connector and proceeded to select all the tables limiting 100 from job_posting_fact to see if i could pull the data and to see that the connection was properly running. As shown in the photos below

`SELECT * FROM job_postings_fact LIMIT 100;`

job_id	company_id	job_title_short	job_title
0	0	Data Analyst	Marketing Data Analyst
55	1	Cloud Engineer	Storage and Virtualization
66	2	Data Analyst	Data Analyst et Scientist
76	3	Data Engineer	Data Engineer
81	4	Data Engineer	Data Engineer
105	5	Data Analyst	Data Analyst
106	6	Data Engineer	Data Lead Engineer (w/)
116	7	Senior Data Engineer	Senior Data Engineer
122	8	Data Analyst	Full Time Data Analyst
134	9	Data Engineer	Data Engineer Remote
139	10	Data Engineer	Lead Data Engineer
145	11	Data Engineer	Data Platform Engineer
148	12	Data Scientist	Data Science Manager
167	13	Data Engineer	Data Engineer

This screen shot displays the two databases in PG Admin PostgresSql showing there is a running connection and that it is up and ready to go.

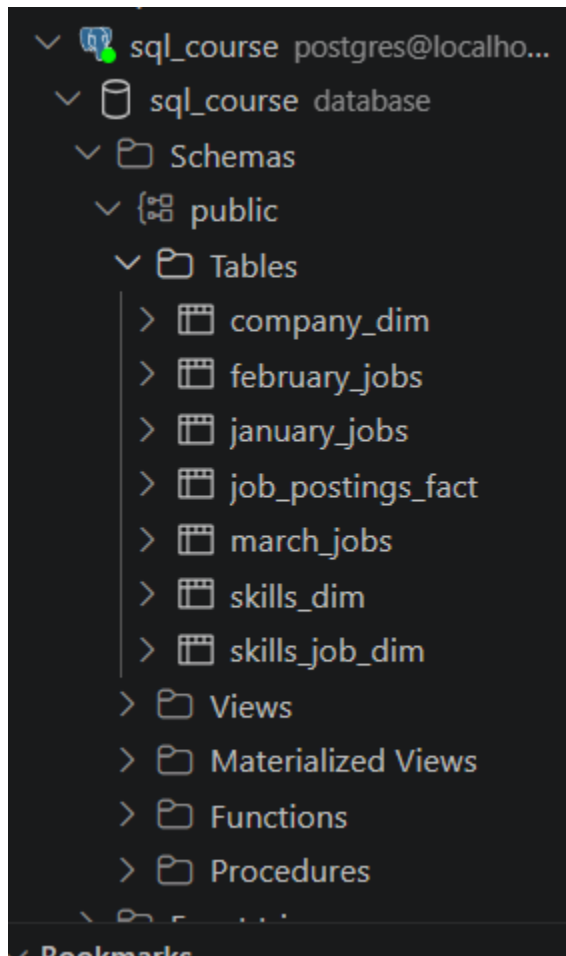


I Created some tables for the first three months in the year where it displays all the information from pulling the from job_posting_fact where each month is aligned to the exact number it is created by

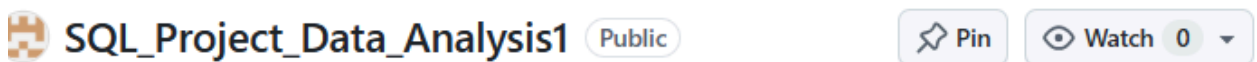
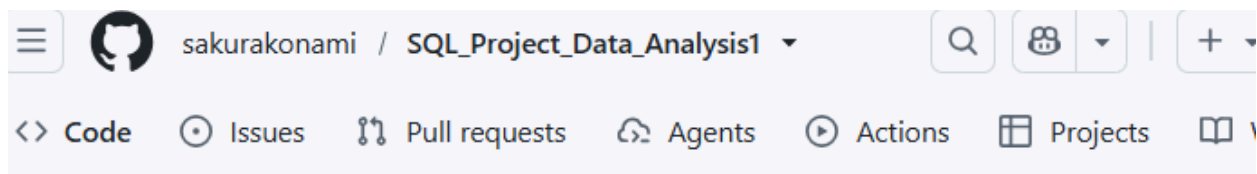
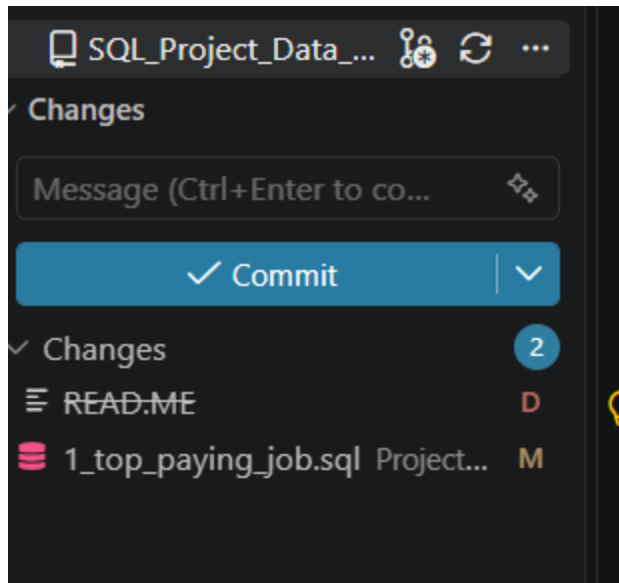
```
3_cases.sql 6
advanced_sql > 3_cases.sql
  ▶ Run on active connection | ≡ Select block
1  -- January
2  CREATE TABLE january_jobs AS
3  SELECT *      ⚠ Evite SELECT * – liste as columnas ne
4  FROM job_postings_fact
5  WHERE EXTRACT(MONTH FROM job_posted_date) = 1;
6
7  -- February
8  CREATE TABLE february_jobs AS
9  SELECT *      ⚠ Evite SELECT * – liste as columnas ne
10 FROM job_postings_fact
11 WHERE EXTRACT(MONTH FROM job_posted_date) = 2;
12
13 -- March
14 CREATE TABLE march_jobs AS
15 SELECT *      ⚠ Evite SELECT * – liste as columnas ne
16 FROM job_postings_fact
17 WHERE EXTRACT(MONTH FROM job_posted_date) = 3;
18
```

As

shown in the left hand column the tables have henceforth been created



Linked project to the Github repository and pushed/ committed the changes to git hub



	sakurakonami	Add initial README with TODO note	778b603 · 21 minutes ago	
	Project_sql	first commit	30 minutes ago	
	sql_load	first commit	30 minutes ago	
	.gitignore	first commit	30 minutes ago	
	READ.ME	first commit	30 minutes ago	
	README.md	Add initial README with TO...	21 minutes ago	



Ab

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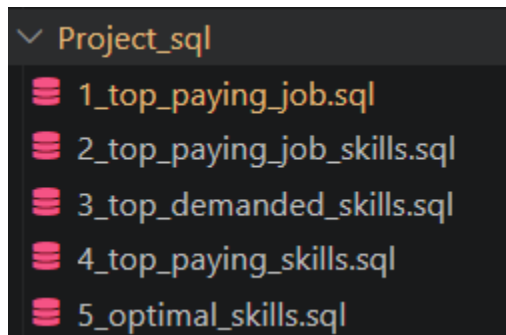
Rel

No
Crea

Created

Sql files based from the following questions i needed to address :

- 1.)What are the top_paying jobs for my role?-1_top_paying_job.sql
- 2.) What are the skills required for these top paying roles?2_top_paying_job_skills.sql
- 3.) What are the most in-demand skills for my role? 3_top_demanded_skills.sql
- 4.) What are the top skills based on salary for my role? 4_top_paying_skills.sql
- 5.) What are the most optimal skills to learn?5_optimal_skills.sql



After creating the files, I proceeded to write SQL code for selecting the job ID, job title, location, schedule type, average yearly salary, and posted date from the `job_postings_fact` table, and also pulled in the company name by aliasing the `name` column from the `company_dim` table, then created a LEFT JOIN on the `company_dim` table to the `job_postings_fact` table matching on the company ID, where the job title is Data Analyst and the location is "Anywhere" and the salary year average is not null, then ordered the salary year average by descending and limited the top 10 results. Here is the code below I created in the file using SQL: The results of the table are shown below, displaying the top 10 paying jobs:

```

▷Run on active connection | ≡ Select block
SELECT
  job_id,
  job_title,
  job_location,
  job_schedule_type,
  salary_year_avg,
  job_posted_date,
  name AS company_name
FROM
  job_postings_fact
LEFT JOIN company_dim ON job_postings_fact.company_id = company_dim.company_id
WHERE
  job_title_short = 'Data Analyst' AND
  job_location = 'Anywhere' AND
  salary_year_avg IS NOT NULL
ORDER BY
  salary_year_avg DESC
LIMIT 10;

```

here is the table results for the top paying jobs as a Data analyst:

job_title	job_location	job_schedule_type	salary_year_avg	job_posted_date	company_name
 Filter...	 Filter...	 Filter...	 Filter...	 Filter...	 Filter...
Data Analyst	Anywhere	Full-time	650000.0	2023-02-20 15:13:33	Mantys
Director of Analytics	Anywhere	Full-time	336500.0	2023-08-23 12:04:42	Meta
Associate Director- Data Insights	Anywhere	Full-time	255829.5	2023-06-18 16:03:12	AT&T
Data Analyst, Marketing	Anywhere	Full-time	232423.0	2023-12-05 20:00:40	Pinterest Job Adve
Data Analyst (Hybrid/Remote)	Anywhere	Full-time	217000.0	2023-01-17 00:17:23	Uclahealthcareers
Principal Data Analyst (Remote)	Anywhere	Full-time	205000.0	2023-08-09 11:00:01	SmartAsset
Director, Data Analyst - HYBRID	Anywhere	Full-time	189309.0	2023-12-07 15:00:13	Inclusively
Principal Data Analyst, AV Performance...	Anywhere	Full-time	189000.0	2023-01-05 00:00:25	Motional
Principal Data Analyst	Anywhere	Full-time	186000.0	2023-07-11 16:00:05	SmartAsset
ERM Data Analyst	Anywhere	Full-time	184000.0	2023-06-09 08:01:04	Get It Recruit - Inf

I proceeded to Push and commit results to Git Hub after completion.

After filtering the top 10 highest-paying Data

Analyst roles in my previous query, I proceeded to write a second SQL query

using a CTE (Common Table Expression) named `top-paying_jobs` that

reused the same logic—selecting job ID, job title, salary year average, and

company name from the `job_postings_fact` table with a LEFT JOIN on

`company_dim`, filtered for Data Scientist roles located "Anywhere" with a

non-null salary, ordered by salary descending and limited to the top 10. I then wrote a main SELECT statement that pulled all columns from the CTE and joined in the `skills` column by creating an INNER JOIN between `top_paying_jobs` and `skills_job_dim` on job ID, followed by another INNER JOIN between `skills_job_dim` and `skills_dim` on skill ID, ordering the final results by salary year average descending. The results displayed a breakdown of which specific skills are required across those top 10 paying jobs, with the key findings showing that Python and SQL lead the list appearing in 4 of the 7 job postings (57%), followed by AWS and Java at 3 postings each (43%), then GCP, PyTorch, Spark, and TensorFlow at 2 postings each (29%), while the remaining 10 skills appeared in only one posting each—suggesting that Python and SQL are common foundations in this group of high-paying positions, while cloud platforms and specialized machine learning tools add further technical breadth.

Here is the SQL code Below :

```
WITH top_paying_jobs AS (  
    SELECT  
        job_id,  
        job_title,  
        salary_year_avg,  
        name AS company_name  
    FROM  
        job_postings_fact  
    LEFT JOIN company_dim ON job_postings_fact.company_id = company_dim.company_id  
    WHERE  
        job_title_short = 'Data Scientist' AND  
        job_location = 'Anywhere' AND  
        salary_year_avg IS NOT NULL  
    ORDER BY  
        salary_year_avg DESC  
    LIMIT 10  
)  
  
SELECT  
    top_paying_jobs.*,  
    skills  
FROM  
    top_paying_jobs  
INNER JOIN skills_job_dim ON top_paying_jobs.job_id = skills_job_dim.job_id  
INNER JOIN skills_dim ON skills_job_dim.skill_id = skills_dim.skill_id  
ORDER BY  
    salary_year_avg DESC;
```

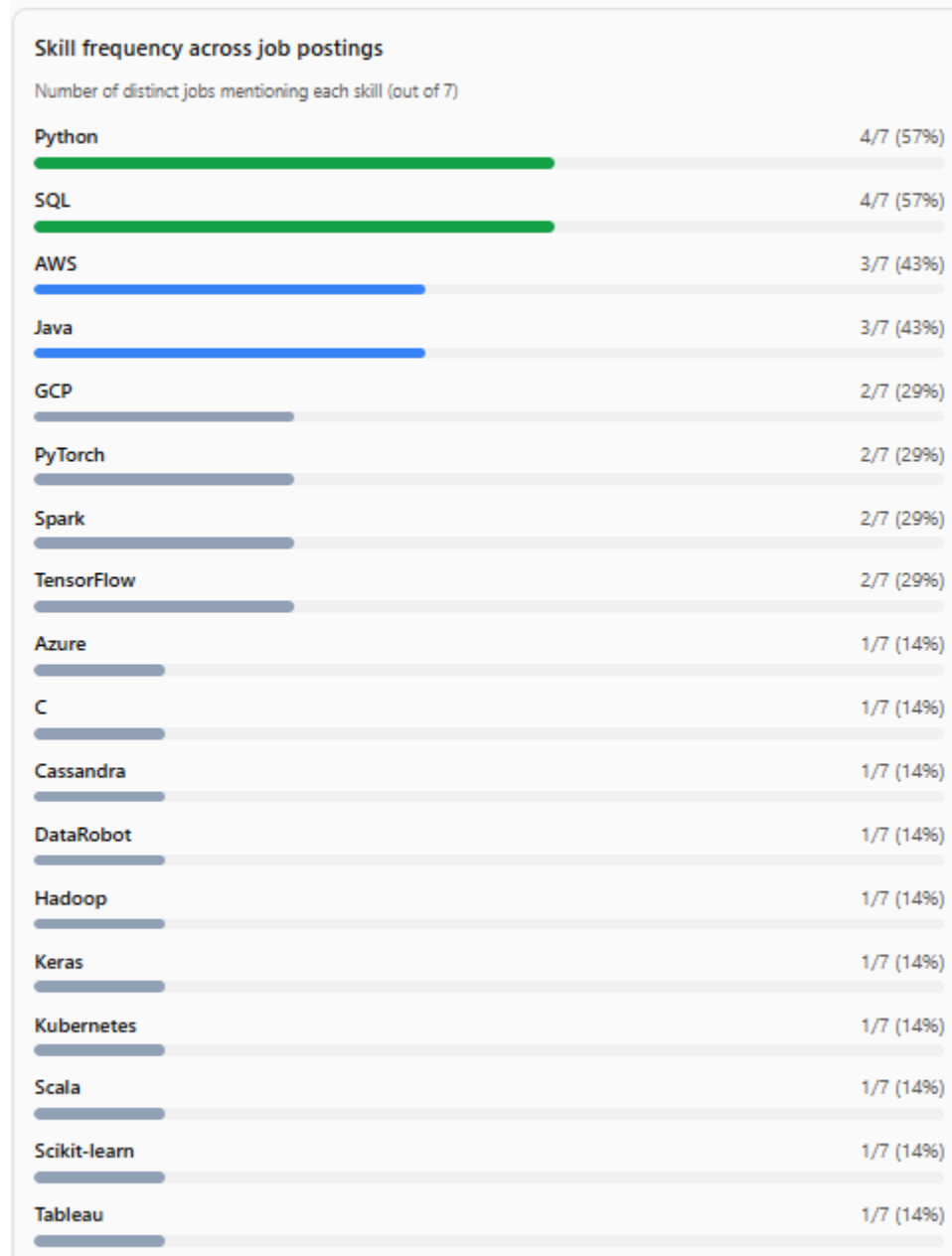
Here were the table results:

job_id	job_title	salary_year_avg	company_name	skills
10145	Staff Data Scientist/Quant Researcher	550000.0	Selby Jennings	sql
10145	Staff Data Scientist/Quant Researcher	550000.0	Selby Jennings	python
1714768	Staff Data Scientist - Business Analytics	525000.0	Selby Jennings	sql
131472	Data Scientist	375000.0	Algo Capital Group	sql
131472	Data Scientist	375000.0	Algo Capital Group	python
131472	Data Scientist	375000.0	Algo Capital Group	java
131472	Data Scientist	375000.0	Algo Capital Group	cassandra
131472	Data Scientist	375000.0	Algo Capital Group	spark
131472	Data Scientist	375000.0	Algo Capital Group	hadoop
131472	Data Scientist	375000.0	Algo Capital Group	tableau
26218	Director Level - Product Management - Data Scie...	320000.0	Teramind	azure
26218	Director Level - Product Management - Data Scie...	320000.0	Teramind	aws
26218	Director Level - Product Management - Data Scie...	320000.0	Teramind	tensorflow
26218	Director Level - Product Management - Data Scie...	320000.0	Teramind	keras
26218	Director Level - Product Management - Data Scie...	320000.0	Teramind	pytorch
26218	Director Level - Product Management - Data Scie...	320000.0	Teramind	scikit-learn
26218	Director Level - Product Management - Data Scie...	320000.0	Teramind	datarobot
26011	Distinguished Data Scientist	300000.0	Walmart	scala
26011	Distinguished Data Scientist	300000.0	Walmart	java
26011	Distinguished Data Scientist	300000.0	Walmart	spark

I then pushed the changed and updates to the Github Link and committed as the message top_paying_job_skills

Here are some of the visualized results

1. Most in-demand skills



I shifted focus to identifying which skills are most in demand for Data Analyst positions. I wrote a SQL query that selects each skill and counts how many job postings require it by using an INNER JOIN between `job_postings_fact` and `skills_job_dim` on job ID, followed by another INNER JOIN between `skills_job_dim` and `skills_dim` on skill ID to retrieve the actual skill names. I then filtered the results to only include Data Analyst roles that offer remote work (`job_work_from_home = True`), grouped the results by skill, ordered them by the demand count in descending order, and limited the output to the top 5 results. The results displayed the five most in-demand skills for remote Data

Analyst positions, revealing which core competencies employers are consistently seeking in the job market. Here is the code below:

```
3_top_demanded_skills.sql
Project_sql > 3_top_demanded_skills.sql
  ▶ Run on active connection | ≡ Select block
1  SELECT
2    skills,
3    COUNT(skills_job_dim.job_id) AS demand_count
4  FROM job_postings_fact
5  INNER JOIN skills_job_dim ON job_postings_fact.job_id = skills_job_dim.job_id
6  INNER JOIN skills_dim ON skills_job_dim.skill_id = skills_dim.skill_id
7  WHERE
8    job_title_short = 'Data Analyst'
9    AND job_work_from_home = True
10 GROUP BY
11   skills
12 ORDER BY
13  demand_count DESC
14 LIMIT 5;
```

Here are the table results showing the top 5 skills ranked highest to lowest:

sql_course: SELECT ski... X		
skills	demand_count	
abc Filter...	abc Filter...	
sql	7291	
excel	4611	
python	4330	
tableau	3745	
power bi	2609	

Then i pushed and committed to github as to_demanded_skills.

After identifying the most in-demand skills, I then analyzed which skills command the highest salaries for Data Analyst roles. I wrote a SQL query that selects each skill along with the average yearly salary associated with it, using the ROUND function to round the average salary to the nearest whole number for cleaner readability. I joined job_postings_fact to skills_job_dim on job ID and then to skills_dim on skill ID using INNER JOINS to link each job posting to its required skills. I filtered the results to only include Data Analyst roles that are remote (job_work_from_home = True) and have a non-null salary value, then grouped the results by skill, ordered them by average salary in

descending order, and limited the output to the top 25 results. The results displayed the 25 highest-paying skills for remote Data Analyst positions, revealing which specialized technical skills correlate with the highest earning potential in the field. Here below is a screenshot of the code i applied in this file to pull the necessary data needed in order to see the table for top paying skills:

```
Run on active connection | Select block
SELECT
    skills,
    ROUND(AVG(salary_year_avg), 0) AS avg_salary
FROM job_postings_fact
INNER JOIN skills_job_dim ON job_postings_fact.job_id = skills_job_dim.job_id
INNER JOIN skills_dim ON skills_job_dim.skill_id = skills_dim.skill_id
WHERE
    job_title_short = 'Data Analyst'
    AND salary_year_avg IS NOT NULL
    AND job_work_from_home = True
GROUP BY
    skills
ORDER BY
    ⚡ avg_salary DESC
LIMIT 25;
```

Here is the data table as a result from my SQL code that pulls that specified data:

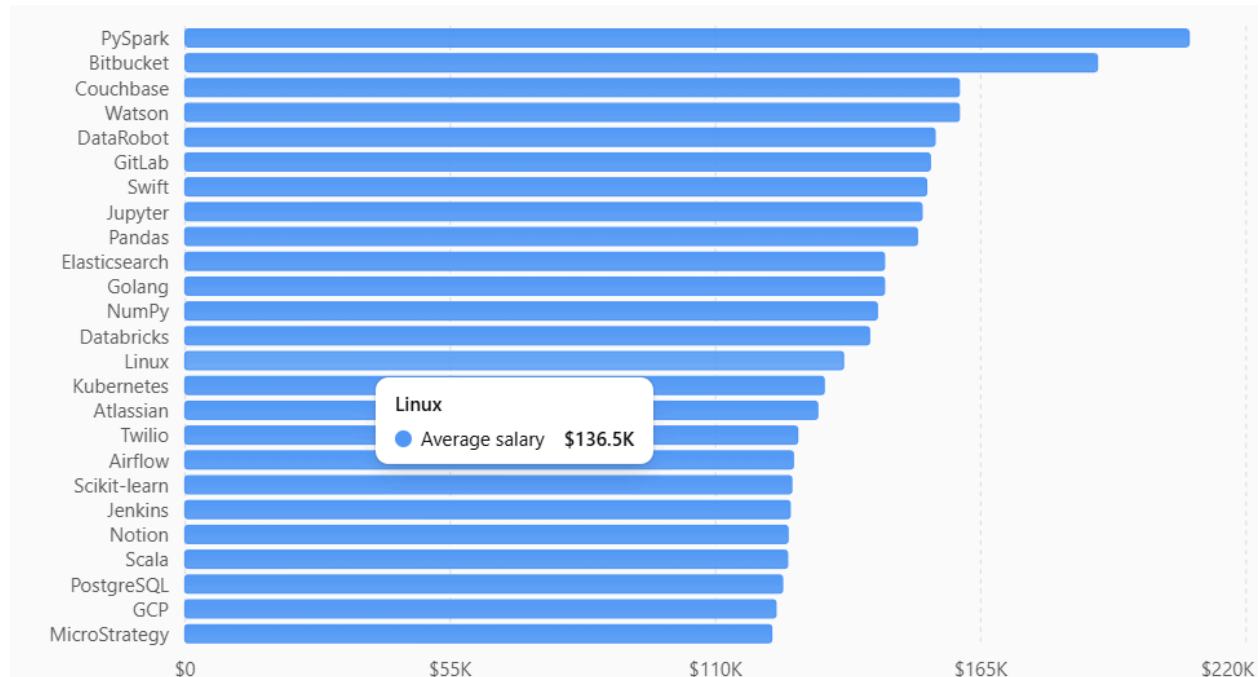
skills	avg_salary
abc Filter...	abc Filter...
pyspark	208172
bitbucket	189155
couchbase	160515
watson	160515
datarobot	155486
gitlab	154500
swift	153750
jupyter	152777
pandas	151821
elasticsearch	145000
golang	145000
numpy	143513
databricks	141907
linux	136508
kubernetes	132500
atlassian	131162
twilio	127000
airflow	126103
scikit-learn	125781
jenkins	125436
notion	125000
scala	124903
postgresql	123879
qcp	122500

Analysis of the Top 25 Highest-Paying Data Analyst Skills

The data reveals an interesting pattern: the highest-paying skills are not limited to traditional data analysis tools. Instead, the list highlights the value of big data processing, cloud infrastructure, automation, and specialized technical platforms.

PySpark leads the ranking with an average salary of \$208,172, followed by Bitbucket at \$189,155. These results suggest that the highest-paying opportunities in this dataset are associated with technical capabilities that extend beyond basic reporting and visualization.

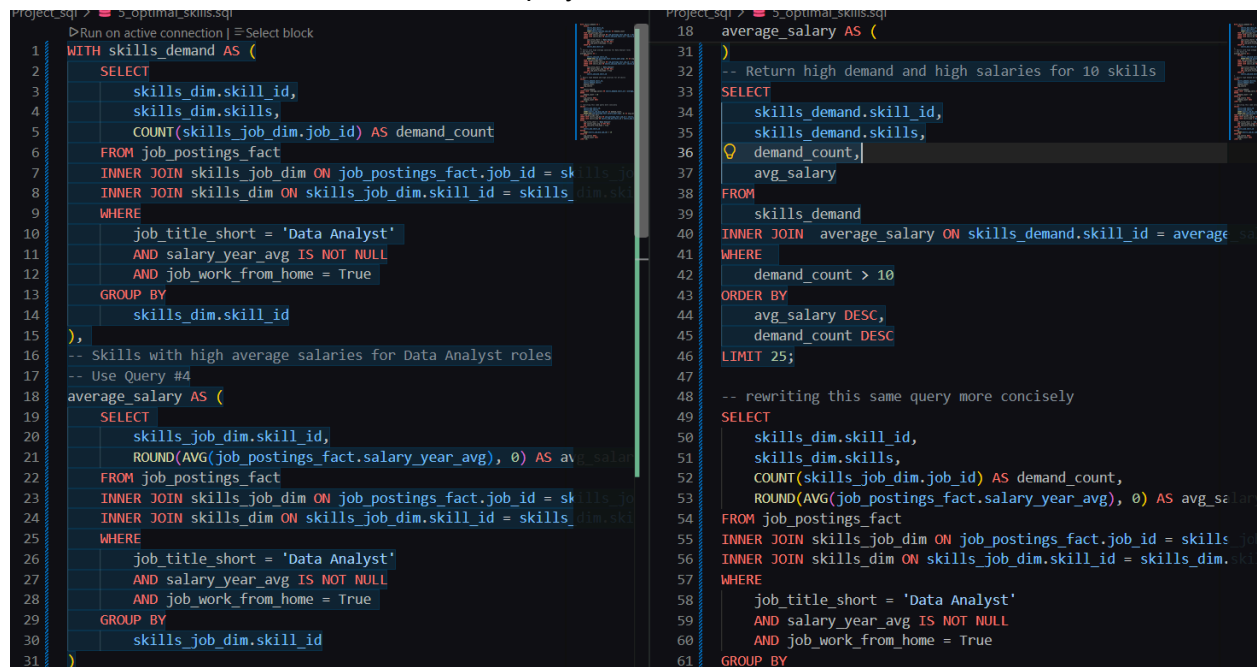
The following analysis breaks down the salary rankings, identifies key trends, and highlights what these findings mean for someone pursuing a career in data analytics.



Key observations

- PySpark is the clear leader, with an average salary of \$208,172, approximately \$19,000 higher than the second-ranked skill.
- Bitbucket is the only other skill exceeding \$180,000.
- Couchbase and Watson share third place at \$160,515.
- The majority of the listed skills fall between \$120,000 and \$160,000.
- Even the lowest-ranked skill, MicroStrategy, is associated with an average salary above \$120,000.

After completing the individual demand and salary analyses, I combined both insights into a single comprehensive query to identify the optimal skills—those that are both highly demanded and highly paid for remote Data Analyst roles. I first wrote the query using two CTEs (Common Table Expressions): `skills_demand` to calculate the demand count for each skill and `average_salary` to calculate the average yearly salary per skill, both filtered for Data Analyst roles that are remote and have a non-null salary. I then joined these two CTEs on skill ID in the final SELECT, filtered for skills with a demand count greater than 10 to ensure statistical relevance, ordered the results by average salary descending and demand count descending, and limited the output to 25 results. I also rewrote the same query more concisely using a single SELECT statement with GROUP BY and a HAVING clause to filter demand count directly, demonstrating my ability to approach the same problem with both a modular CTE-based approach and a streamlined single-query approach. The results displayed the top 25 skills that strike the best balance between market demand and salary potential, providing actionable insights into which skills offer the strongest career value for remote Data Analysts. Here is the SQL code I dived into below to display the outcode for the needed table and data:



```

1  WITH skills_demand AS (
2      SELECT
3          skills_dim.skill_id,
4          skills_dim.skills,
5          COUNT(skills_job_dim.job_id) AS demand_count
6      FROM job_postings_fact
7      INNER JOIN skills_job_dim ON job_postings_fact.job_id = skills_job_dim.job_id
8      INNER JOIN skills_dim ON skills_job_dim.skill_id = skills_dim.skill_id
9      WHERE
10         job_title_short = 'Data Analyst'
11         AND salary_year_avg IS NOT NULL
12         AND job_work_from_home = True
13      GROUP BY
14         skills_dim.skill_id
15  ),
16  -- Skills with high average salaries for Data Analyst roles
17  -- Use Query #4
18  average_salary AS (
19      SELECT
20          skills_job_dim.skill_id,
21          ROUND(AVG(job_postings_fact.salary_year_avg), 0) AS avg_salary
22      FROM job_postings_fact
23      INNER JOIN skills_job_dim ON job_postings_fact.job_id = skills_job_dim.job_id
24      INNER JOIN skills_dim ON skills_job_dim.skill_id = skills_dim.skill_id
25      WHERE
26         job_title_short = 'Data Analyst'
27         AND salary_year_avg IS NOT NULL
28         AND job_work_from_home = True
29      GROUP BY
30         skills_job_dim.skill_id
31  )
32  SELECT
33      skills_demand.skill_id,
34      skills_demand.skills,
35      demand_count,
36      avg_salary
37  FROM
38      skills_demand
39  INNER JOIN average_salary ON skills_demand.skill_id = average_salary.skill_id
40  WHERE
41      demand_count > 10
42  ORDER BY
43      avg_salary DESC,
44      demand_count DESC
45  LIMIT 25;
46
47  -- rewriting this same query more concisely
48  SELECT
49      skills_dim.skill_id,
50      skills_dim.skills,
51      COUNT(skills_job_dim.job_id) AS demand_count,
52      ROUND(AVG(job_postings_fact.salary_year_avg), 0) AS avg_salary
53  FROM job_postings_fact
54  INNER JOIN skills_job_dim ON job_postings_fact.job_id = skills_job_dim.job_id
55  INNER JOIN skills_dim ON skills_job_dim.skill_id = skills_dim.skill_id
56  WHERE
57      job_title_short = 'Data Analyst'
58      AND salary_year_avg IS NOT NULL
59      AND job_work_from_home = True
60  GROUP BY
61      skills_dim.skill_id
62  ORDER BY
63      avg_salary DESC,
64      demand_count DESC
65  LIMIT 25;

```

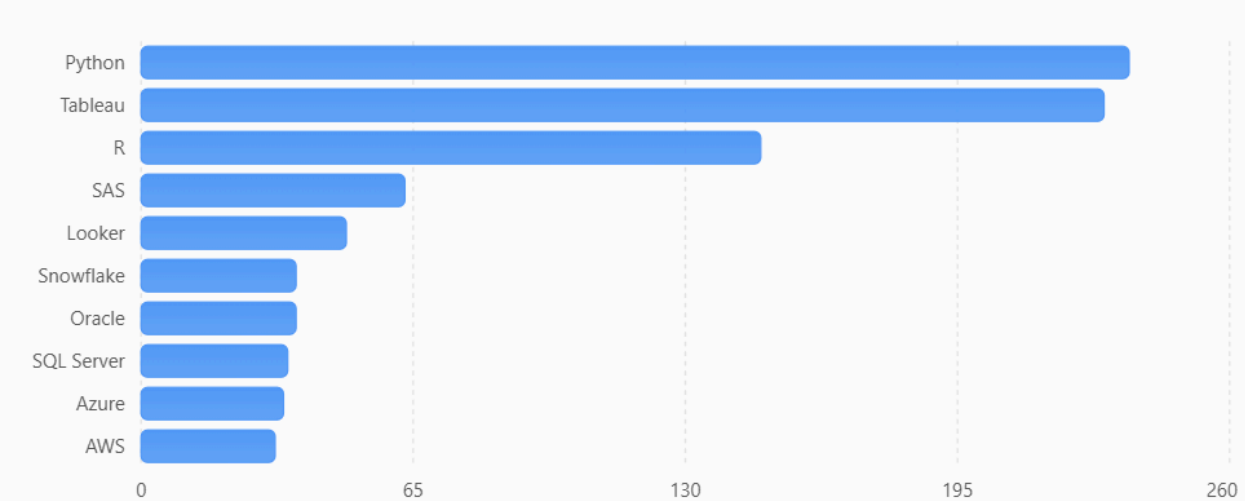
The data pulled from the files in the table shown below :

skill_id	skills	demand_count	avg_salary
abc Filter...	abc Filter...	abc Filter...	abc Filter...
8	go	27	115320
234	confluence	11	114210
97	hadoop	22	113193
80	snowflake	37	112948
74	azure	34	111225
77	bigquery	13	109654
76	aws	32	108317
4	java	17	106906
194	ssis	12	106683
233	jira	20	104918
79	oracle	37	104534
185	looker	49	103795
2	nosql	13	101414
1	python	236	101397
5	r	148	100499
78	redshift	16	99936
187	qlik	13	99631
182	tableau	230	99288
197	ssrs	14	99171
92	spark	13	99077

I proceed to commit each change and push into the gitHub repository. Here is a visual of the data for the top optimal skills

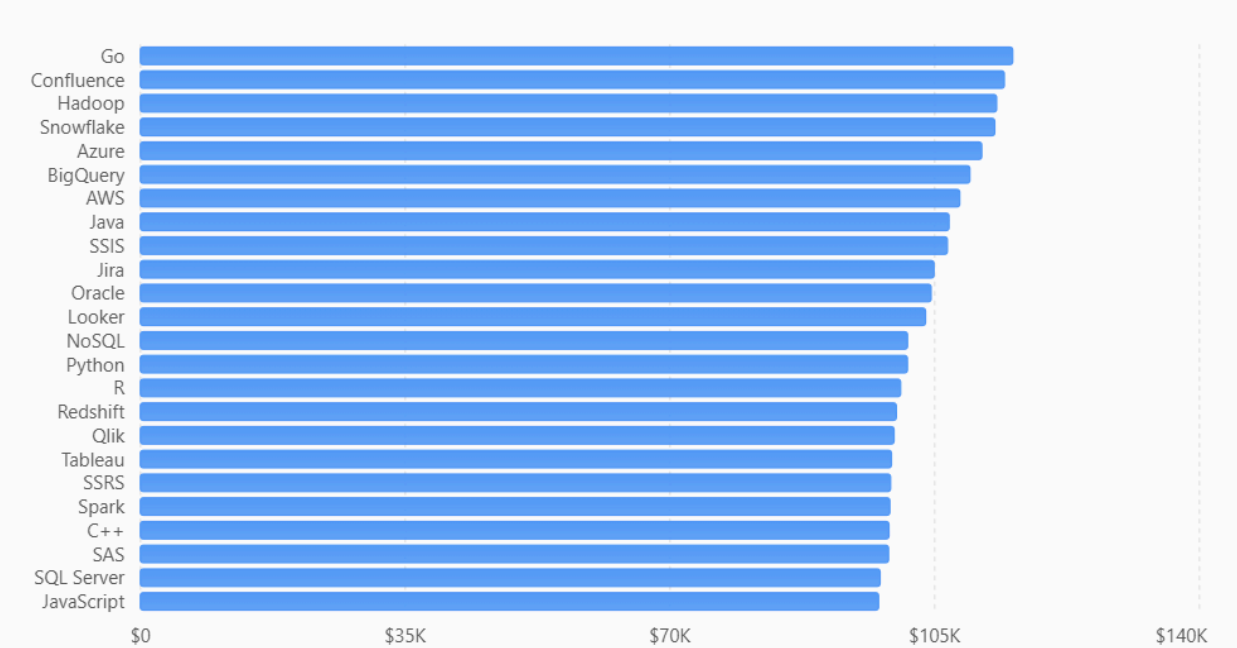
Top 10 skills by demand

Number of job postings mentioning each skill.



Average salary across the 25 listed skills

Annual average salary in USD, ranked from highest to lowest.

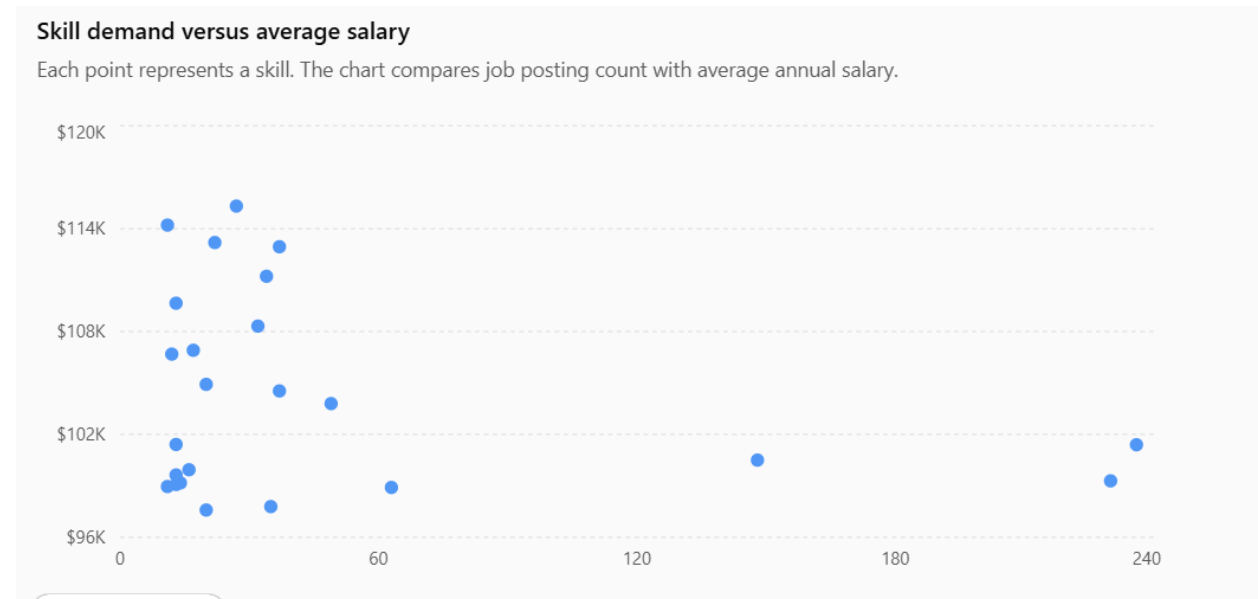


What the comparison tells us

The chart highlights three distinct patterns:

1. High demand, moderate salary: Python and Tableau are frequently requested, but their average salaries are below those of several specialized technologies.
2. Lower demand, higher salary: Go, Hadoop, and BigQuery have relatively fewer postings but higher average salaries.
3. High demand, competitive salary: Snowflake, Azure, and AWS combine substantial demand with average salaries above \$108,000.

This suggests that salary and demand do not move in lockstep. However, the data alone cannot establish whether specialization, seniority, industry, or other factors explain the differences.



5. Five major takeaways

1. Python and Tableau are the most frequently requested skills. Their demand counts of 236 and 230 make them prominent foundations in this dataset.
2. High salary does not necessarily mean high demand. Go ranks first in average salary but appears in only 27 postings, compared with Python's 236.
3. Cloud technologies combine demand and competitive compensation. Snowflake, Azure, and AWS are represented in more than 30 postings each, with average salaries exceeding \$108,000.
4. Data analyst roles encompass a broad technical skill set. Programming, visualization, cloud computing, data warehousing, and big data processing all feature in the results.
5. The most useful career insights come from considering salary and demand together. A skill's average salary alone does not show how frequently employers request it or how many opportunities are available.